Docker introduction

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Docker is an open-source platform used to develop, deploy and run applications. Through container technology, developers can create consistent, portable and scalable application environments, which improves development efficiency and application reliability.

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Pulling Docker images online may not be possible in China at present, the tutorial is for demonstration only
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1. Containers

A Docker container is a lightweight, standalone executable software package that contains everything needed to run an application: the code, the runtime, system tools, system libraries and settings.

Container characteristic	Description
Lightweight	Containers share the kernel of the host operating system
Independent	Containers are isolated from each other, with their own file systems, process spaces and network interfaces
Portable	Containers can run on any platform that supports Docker

Note: Docker containers are built on top of the underlying operating system. The instruction sets differ between CPU architectures, so containers built for different architectures may run into compatibility problems.

For example, an image that only has a `linux/amd64` version cannot run directly on an ARM64 host. You should confirm the architectures it supports before pulling an image; some hosts can also be configured with emulators such as QEMU, but performance and compatibility may be affected. Multi-architecture images usually have Docker automatically select the version that matches the host.

2. Images

An image is a read-only template used to create containers.

Image characteristic	Description
Immutable	Images are read-only and never change once they are created

Image characteristic	Description
Layered structure	An image consists of several layers, each representing a state of the image

Note: Changes produced while a container runs are by default stored in that container's writable layer, and they are kept when the same container is started again after being stopped; deleting the container loses that data. When these changes need to be reused in a new container, they can be committed as a new image. Data in mounted volumes or bind-mounted directories is stored independently of the image and the container writable layer.

3. Docker Engine

The Docker Engine is a client-server application that includes the Docker daemon, the REST API and the command-line interface (CLI).

3.1. Docker daemon

Responsible for building, running and managing containers: it runs as a background service, starts when the host boots, keeps running in the background and listens for Docker API requests.

3.2. Docker CLI

Provides the command-line tool used to interact with the Docker daemon.

4. Docker Hub

Docker Hub is a cloud-based public registry used to store and distribute Docker images.

5. Main advantages of Docker

Docker is a powerful tool that has changed the way applications are developed, deployed and run.

Main advantage	Description
Consistency	Applications in containers always run in the same environment
Isolation	Containers are isolated from each other, ensuring application security and stability
Portability	Cross-platform and easy to migrate
Efficiency	Shares the operating system kernel, making it lighter than traditional virtual machines and consuming fewer resources